

ClearGS: Reliability-Aware Gaussian Splatting from Handheld Videos

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Abstract

We present ClearGS for 3D Gaussian Splatting (3DGS) from handheld videos with uneven viewpoint coverage and mixed frame quality. Rather than selecting frames with binary decisions, ClearGS uses Reliability-aware View Allocation (RVA) to assign graded raw-supervision weights based on appearance reliability, degradation risk, and geometric utility, while weakly reactivating useful suppressed frames to maintain trajectory coverage. Since weighting cannot restore details lost to blur or distortion, ClearGS further introduces Render-Guided In-Video Restoration (RIVR). The current 3DGS render provides a pose-aligned structural candidate, a frozen no-reference restoration expert restores the corresponding raw video observation without any clean reference image, and no-reference perceptual scores select among the render, restored observation, and high-frequency fused candidate. ClearGS then applies Full-Trajectory Repair Consolidation to revisit accepted repairs and preserve details introduced early. On GS2E and GSOTM, ClearGS achieves state-of-the-art overall performance, with consistent CLIP-IQA and MUSIQ gains and LPIPS reductions in most degradation settings, without paired sharp supervision or matched clean references.

Introduction

Recent advances in 3D Gaussian Splatting (3DGS) enable efficient scene reconstruction and high-quality novel-view rendering (Kerbl et al. 2023), but they assume input views with reasonably consistent appearance and geometry. Handheld videos often break this assumption: frame quality changes along the trajectory, and viewpoint coverage is uneven. Training all frames with equal strength transfers blur, distortion, and inconsistent appearance into the scene. Existing frame- and view-selection methods retain or acquire views according to quality, coverage, uncertainty, or information gain (Polyzos et al. 2025; Xue et al. 2026; Chen et al. 2026b), but binary keep-or-drop decisions are too coarse for handheld video. A degraded frame may still provide useful geometric evidence; dropping it weakens coverage, while using it at full weight contaminates appearance. We therefore formulate handheld-video 3DGS as a mixed-quality supervision problem that separates appearance reliability from geometric utility.

ClearGS first addresses this problem with Reliability-aware View Allocation (RVA). For each frame, RVA estimates appearance reliability, degradation risk, and geometric

utility, then assigns an ordered raw-supervision weight. Reliable low-risk frames receive strong supervision, geometrically useful observations receive attenuated supervision, and unreliable low-utility frames are suppressed. To avoid coverage holes, RVA also weakly reactivates useful suppressed frames in under-covered trajectory regions, preserving viewpoint coverage without letting degraded appearance dominate the initial reconstruction.

Weighted supervision alone cannot recover details already lost to blur or distortion. Existing restoration–distillation methods often depend on nearby RGB references, aggregated sparse-view features, or depth-guided correspondences (Wu et al. 2025a; Liu, Zhou, and Huang 2024; Wu et al. 2025b; Yin et al. 2025; Cao et al. 2026). These cues are unreliable in handheld videos: neighboring frames may suffer from the same degradation, and corrupted appearance can be propagated across views. Render-Guided In-Video Restoration (RIVR) extends ClearGS from raw supervision to repaired evidence. For each target pose, RIVR renders the current scene as a pose-aligned structural candidate and restores the corresponding raw video observation with a frozen no-reference restoration expert, without using any clean reference image or paired sharp target. RIVR also constructs a detail-fused candidate by combining the rendered base with restored high-frequency details. It then selects among the render, restored observation, and fused candidate using no-reference perceptual scores, and appends the selected result as a pseudo observation for later distillation.

Progressive distillation can weaken details introduced by repairs accepted early in the trajectory. ClearGS therefore performs Full-Trajectory Repair Consolidation, which reuses the latest accepted repair at each target pose without sampling new poses or generating new candidates. Together, RVA, RIVR, and Full-Trajectory Repair Consolidation preserve reliable captured evidence, recover video-grounded details, and maintain weakly observed regions. Experiments on GS2E (Li et al. 2025) and GSOTM (Seiskari et al. 2024) show state-of-the-art overall performance, with sharper boundaries and more stable textures under mixed-quality observations.

Our contributions are threefold:

- We formulate 3DGS reconstruction from handheld videos as a mixed-quality supervision problem that separates appearance reliability, degradation risk, and geometric utility.

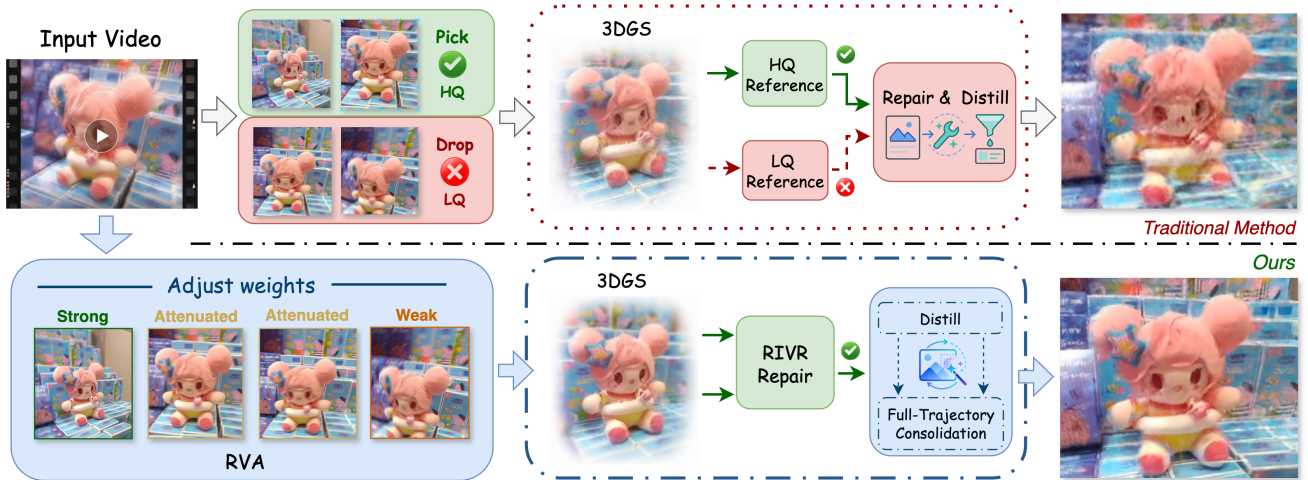


Figure 1: Traditional pipelines either train or select handheld-video frames without graded reliability control, then repair degraded views using nearby references or cross-view cues that may share the same artifacts. ClearGS instead uses RVA to separate appearance reliability from geometric utility, and applies no-reference RIVR to add selected repair evidence without matched clean references.

- We introduce RVA, which assigns graded raw-supervision weights and preserves trajectory coverage through weak reactivation without allowing degraded observations to dominate scene appearance.
- We introduce RIVR, a no-reference in-video restoration and distillation strategy that selects repair evidence without clean references and consolidates accepted repairs through Full-Trajectory Repair Consolidation.

is often to improve acquisition efficiency or expose under-observed geometry. In fixed handheld videos, however, binary selection cannot use the geometric value of a degraded frame without also accepting its unreliable appearance. RVA separates appearance reliability, degradation risk, and geometric utility, assigning graded raw-supervision weights rather than hard decisions. It also weakly reactivates useful suppressed frames to avoid trajectory coverage holes.

Related Work

Scene Reconstruction from Blurry Images and Handheld Videos

Blur-aware radiance-field methods recover sharp scenes by learning spatially varying kernels, imposing physical priors, or estimating exposure-time camera trajectories (Ma et al. 2022; Lee et al. 2023a; Wang et al. 2023; Lee et al. 2023b). Recent 3DGS methods model blur through Gaussian covariance manipulation, per-pixel kernels, or continuous camera motion (Lee et al. 2024; Zhao, Wang, and Liu 2024; Peng et al. 2024; Lee et al. 2025), and handheld-video methods further compensate for rolling shutter and pose errors (Seiskari et al. 2024; Zhang et al. 2026). These methods explain degradation through image-formation or camera-motion models. ClearGS addresses a different failure mode: frame quality and viewpoint coverage vary along a fixed handheld trajectory. It controls the appearance influence of each frame while retaining observations that still provide useful geometric evidence.

Frame and View Selection for Reconstruction

Frame and active view selection methods retain informative observations or acquire new viewpoints according to image quality, coverage, uncertainty, or information gain (Polyzos et al. 2025; Xue et al. 2026; Chen et al. 2026b). Their goal

3DGS Recovery from Imperfect Observations

Methods for sparse or imperfect observations stabilize 3DGS using depth priors, multi-view constraints, structural regularization, and Gaussian control (Chung, Oh, and Lee 2024; Xiong et al. 2023; Zhu et al. 2024; Li et al. 2024; Xu et al. 2024; Zhang et al. 2024; Park, Ryu, and Kim 2025; Chen et al. 2026a; Song et al. 2025). Generative methods repair rendered views or add pseudo-observations (Wu et al. 2024; Liu, Zhou, and Huang 2024; Wu et al. 2025b; Shen et al. 2026; Jiang et al. 2026; Li et al. 2026; Ni et al. 2025; Zhu et al. 2026), while recent 3DGS repair methods often condition restoration on nearby RGB references, sparse-view features, or depth-guided correspondences (Wu et al. 2025a; Yin et al. 2025; Cao et al. 2026). These cues can be unreliable in mixed-quality handheld videos, where neighboring frames may share the same degradation and corrupted appearance can propagate across views. RIVR instead restores the corresponding raw video observation with a frozen no-reference restoration expert, using no clean reference image or paired sharp target. The current scene render provides a pose-aligned structural candidate for selection and fusion, and accepted repairs are later reused through Full-Trajectory Repair Consolidation.

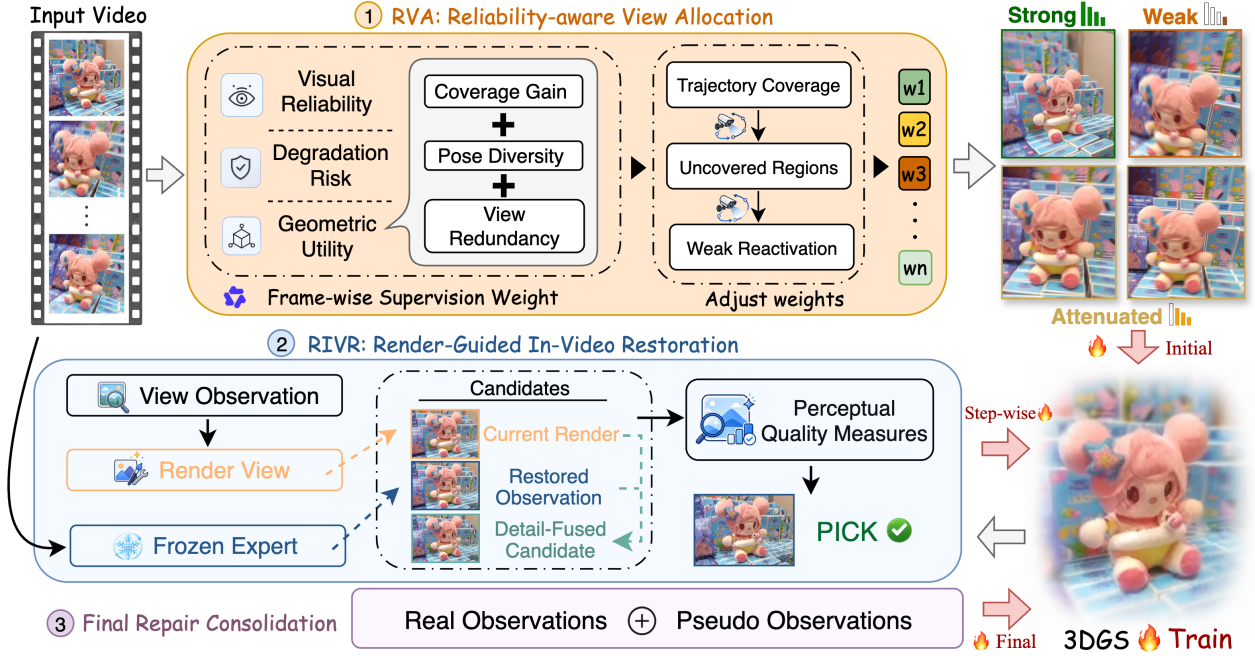


Figure 2: Overview of ClearGS. RVA assigns frame-wise raw-supervision weights from appearance reliability, degradation risk, and geometric utility, and weakly reactivates useful suppressed frames in uncovered trajectory regions. RIVR selects among the current render, restored raw observation, and detail-fused candidate using no-reference quality scores, then distills accepted pseudo observations with real observations.

Method

Problem Definition

We consider a handheld posed video sequence

$$\mathcal{V} = \{(\mathbf{I}_t, \boldsymbol{\pi}_t)\}_{t=1}^T, \quad (1)$$

where $\mathbf{I}_t$ is an input video frame and $\boldsymbol{\pi}_t$ denotes its camera parameters. Such videos contain observations of heterogeneous quality: clear frames coexist with frames affected by motion blur, defocus, distortion, or other degradations. Moreover, viewpoint coverage is often uneven along the camera trajectory. A degraded frame may therefore be unsafe as appearance supervision while still providing the only observation of a weakly covered region.

Our goal is to reconstruct a 3DGS scene $\mathcal{G}^*$ from $\mathcal{V}$ that faithfully renders both observed and novel views. During reconstruction, only the input video is available; paired sharp targets and high-quality reference images matched to the target pose are unavailable. The central challenge is to preserve the geometric coverage offered by degraded observations without directly writing their unreliable appearance into the scene representation.

Overview

ClearGS follows a reliability-first principle for handheld video reconstruction: we first establish a reliable initial scene before introducing repaired observations. RVA evaluates frames from appearance and geometric perspectives, allocates reliability-aware supervision, and preserves necessary trajectory coverage to initialize $\mathcal{G}_0$. Starting from $\mathcal{G}_0$,

RIVR forms three target-pose candidates: the current 3DGS render, the restored raw video frame, and a fused result that combines the rendered base with restored high-frequency details. The selected candidate is appended as an additional pseudo observation and optimized together with the original video supervision, complementing rather than replacing the captured frames. Finally, full-trajectory repair consolidation gathers the accepted pseudo observations across the trajectory and performs a final joint distillation pass, producing the optimized scene $\mathcal{G}^*$.

Reliability-aware View Allocation (RVA)

Handheld video reconstruction is influenced by both the appearance reliability of individual frames and their geometric utility for scene reconstruction. Frames with reliable appearance provide stronger photometric supervision, while frames captured from complementary viewpoints provide additional geometric constraints. Therefore, ClearGS evaluates each frame from both appearance and geometric perspectives and assigns reliability-aware supervision weights, instead of simply retaining or discarding frames based on a single criterion (Polyzos et al. 2025; Xue et al. 2026; Chen et al. 2026b).

Appearance Evidence We first evaluate whether a frame provides reliable appearance information. For each input frame $\mathbf{I}_i$, RVA uses a frozen Qwen3-VL (Bai et al. 2025) as an image-based evaluator. Qwen3-VL predicts appearance quality cues and degradation patterns, which are aggregated into two complementary scores. Quality-related cues, including sharpness, structural clarity, texture reliability, artifact clean-

Table 1: RVA notation, parameter settings, and supervision policy. The standard supervision conditions are evaluated from top to bottom. Afterward, trajectory coverage compensation may reactivate a suppressed frame in an uncovered trajectory bin with a fixed weak-supervision weight.

Part	Quantity / role	Definition or condition	Parameters / weight
Appearance	$\mathbf{r}_i$	ordered quality cues: sharpness, structural clarity, texture reliability, artifact cleanliness, exposure quality	$\mathbf{w}_R = (0.30, 0.25, 0.20, 0.15, 0.10)$
	$\mathbf{b}_i$	ordered degradation cues: motion blur, defocus, ghosting, exposure failure, unreliable texture, artifacts	$\mathbf{w}_B = (0.25, 0.15, 0.20, 0.15, 0.15, 0.10)$
	R_i	appearance reliability score	$R_i = \mathbf{w}_R^\top \mathbf{r}_i$
	B_i	degradation risk score	$B_i = \mathbf{w}_B^\top \mathbf{b}_i$
	$\mathcal{A}$	appearance-reliable frame set	$\tau_R = 0.70, \tau_B = 0.60$
Geometry	d_{ij}^c	normalized camera-center distance between frames i and j	Euclidean distance normalized to $[0, 1]$
	d_{ij}^v	normalized viewing-direction difference between frames i and j	viewing-direction angle normalized to $[0, 1]$
	D_i	$D_i = \max_{j \in \mathcal{A}} (\gamma_c d_{ij}^c + \gamma_v d_{ij}^v)$	$(\gamma_c, \gamma_v) = (0.65, 0.35)$
	O_i	$O_i = \max_{j \in \mathcal{A}} \frac{ \mathcal{P}_i \cap \mathcal{P}_j }{ \mathcal{P}_i \cup \mathcal{P}_j }$	—
	G_i	geometric utility score	$(\alpha, \beta, \eta) = (0.45, 0.35, 0.30)$
Trajectory compensation	Selection coefficients	coefficients in Eq. (6)	$(\lambda_G, \lambda_R, \lambda_B, \lambda_Z) = (0.35, 0.25, 0.30, 0.10)$
	Reactivated supervision	suppressed frame selected as i_m^* in an uncovered trajectory bin	$w_{i_m^*} = 0.12$
Supervision assignment	Strong	$R_i \geq 0.70, B_i \leq 0.60$	$w_i = 0.80 + 0.20R_i$
	Geometrically attenuated	$G_i \geq 0.60, B_i \leq 0.80$	$w_i = 0.30 + 0.30(0.35R_i + 0.65G_i)$
	Auxiliary attenuated	$R_i \geq 0.35, G_i \geq 0.25, B_i \leq 0.75$	$w_i = 0.30 + 0.30(0.50R_i + 0.50G_i)$
	Coverage-reactivated	initially suppressed but selected as i_m^*	$w_i = 0.12$
	Suppressed	otherwise and not selected for coverage compensation	$w_i = 0$

liness, and exposure quality, are combined into a reliability score R_i , indicating whether the frame can provide trustworthy appearance supervision. Degradation patterns, including motion blur, defocus, ghosting, exposure failure, unreliable texture, and artifacts, are combined into a degradation risk score B_i , indicating the likelihood of introducing unreliable appearance into the reconstruction. Frames with high R_i and low B_i are considered appearance-reliable frames $\mathcal{A}$:

$$\mathcal{A} = \{j \mid R_j \geq \tau_R, B_j \leq \tau_B\}, \quad (2)$$

where τ_R and τ_B are predefined thresholds for reliability and degradation risk, respectively. The score aggregation weights and threshold values are summarized in Table 1.

Geometric Evidence Besides appearance information, ClearGS considers the geometric contribution of each frame to scene reconstruction. A frame with degraded appearance may still provide valuable observations from complementary viewpoints. We estimate geometric utility based on the COLMAP sparse reconstruction and camera poses.

Given the reliable frame set $\mathcal{A}$ obtained from appearance assessment, we construct a reference coverage by merging their visible 3D points:

$$\mathcal{P}_{\text{rel}} = \bigcup_{j \in \mathcal{A}} \mathcal{P}_j, \quad (3)$$

where $\mathcal{P}_j$ denotes the set of COLMAP 3D points observed by frame j . This coverage represents scene regions supported by reliable appearance observations.

For each frame i , we measure geometric utility using three complementary cues: coverage gain C_i , pose diversity D_i , and visibility redundancy O_i . Coverage gain measures additional 3D points observed by frame i beyond $\mathcal{P}_{\text{rel}}$:

$$C_i = \frac{\sum_{p \in \mathcal{P}_i \setminus \mathcal{P}_{\text{rel}}} \frac{1}{\sqrt{n_p}}}{\sum_{p \in \mathcal{P}_i} \frac{1}{\sqrt{n_p}}}, \quad (4)$$

where n_p denotes the number of COLMAP images observing point p , assigning larger weights to sparsely observed points. Pose diversity measures the complementarity of the camera pose of frame i with reliable observations, while visibility redundancy measures its overlap with existing coverage. The geometric utility is computed as:

$$G_i = \text{clip}_{[0,1]} (\alpha C_i + \beta D_i - \eta O_i), \quad (5)$$

where α, β , and η balance the three geometric cues. A higher G_i indicates that the frame provides complementary geometric evidence and receives higher priority during supervision allocation. The formulations of D_i and O_i , together with their corresponding weights, are summarized in Table 1.

Trajectory Coverage Compensation However, independent frame-wise decisions may suppress neighboring frames in the same trajectory region, causing local coverage gaps. To preserve sufficient camera coverage during scene initialization, RVA further performs trajectory coverage compensation. Specifically, the trajectory is partitioned into pose-aware bins according to camera motion. If a bin already contains active frames, it remains unchanged. Otherwise, RVA reactivates one suppressed frame i_m^* :

$$i_m^* = \arg \max_{i \in \mathcal{S}_m} (\lambda_G G_i + \lambda_R R_i + \lambda_B (1 - B_i) + \lambda_Z Z_i), \quad (6)$$

where $\mathcal{S}_m$ denotes the suppressed frames in the uncovered trajectory bin. Z_i denotes the trajectory centrality score, where larger values favor frames closer to the bin center. The trajectory-compensation coefficients and weak reactivation weight are summarized in Table 1.

Finally, RVA assigns each frame a supervision role according to Table 1. The resulting supervision set, including both allocated frames and trajectory-compensated frames, is used to initialize $\mathcal{G}_0$.

Table 2: RIVR candidates for no-reference in-video restoration.

Candidate	Construction	Role
Render	$\mathbf{R}_t$	Pose-aligned structural candidate
Restored	$\mathbf{D}_t$	Restoration of the raw target observation
Fused	$\mathbf{F}_t = \mathbf{R}_t + \mathbf{D}_t - \mathcal{B}_\sigma(\mathbf{D}_t)$	Render base with restored high-frequency detail

Render-Guided In-Video Restoration (RIVR)

After RVA builds the initial scene $\mathcal{G}_0$, some details from degraded frames may still be missing or weak in the reconstruction. RIVR addresses this problem with no-reference in-video restoration. It does not require a matched clean image, a paired sharp target, nearby RGB references, aggregated cross-view features, or explicit geometric correspondences (Wu et al. 2025a; Liu, Zhou, and Huang 2024; Wu et al. 2025b; Yin et al. 2025; Cao et al. 2026). Instead, it uses two signals available at the same target pose: the current 3DGS render and the corresponding raw video observation.

For each target frame t , RIVR renders the current scene as $\mathbf{R}_t$ and retrieves the raw observation $\mathbf{O}_t$. The raw observation, rather than the rendered image, is processed by a frozen Restormer motion-deblurring expert (Zamir et al. 2022), producing $\mathbf{D}_t$. The render provides pose-aligned scene structure, but it is not treated as a clean reference. The restored observation provides target-view appearance evidence recovered from the original video frame. RIVR then forms the candidate set summarized in Table 2.

Because clean references are unavailable during reconstruction, RIVR selects among the candidates with a reference-free rule. MUSIQ (Ke et al. 2021) is used as the primary score because it responds to blur, detail loss, and unnatural local structure, while CLIP-IQA is used only when MUSIQ scores are close. The selected candidate is appended as a pseudo observation for later distillation. Thus, restoration evidence is used only after being chosen from render, restored, and fused candidates, without assuming any clean target image or cross-view reference.

Joint Distillation

Following restoration-guided reconstruction methods (Wu et al. 2025a; Yin et al. 2025; Cao et al. 2026), ClearGS distills accepted RIVR outputs with the active real views selected by RVA. At round r , the training set is

$$\mathcal{D}^{(r)} = \mathcal{D}_{\text{real}} \cup \bigcup_{\ell=1}^r \mathcal{D}_{\text{pseudo}}^{(\ell)}. \quad (7)$$

Repaired views are appended rather than used to replace real observations, keeping captured video evidence while adding selected restoration details.

Since later rounds may weaken repairs accepted early, ClearGS applies Full-Trajectory Repair Consolidation after all frames are processed. It reuses the accepted repairs from the whole trajectory, keeps only the latest repair for repeated target frames, and introduces no new poses or repair candidates.

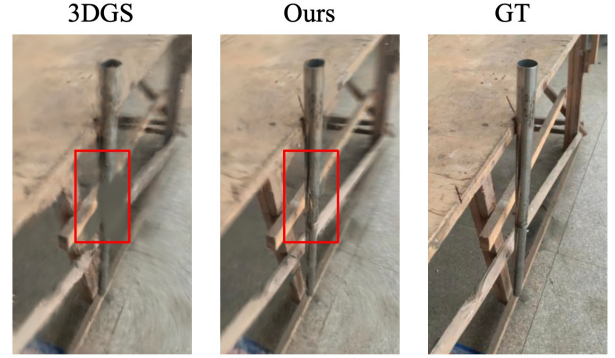


Figure 3: Qualitative comparison of renderings at observed input camera poses produced by all-view 3DGS and the Select-view initialization of ClearGS. Red boxes highlight regions where our reconstruction better preserves thin structures, object boundaries, and local appearance relative to all-view supervision.

Experiment

Experimental Setup

We evaluate ClearGS on GS2E (Li et al. 2025) and Gaussian Splatting on the Move (GSOTM) (Seiskari et al. 2024). GS2E contains ten scenes at each of three synthetic blur levels: slight, medium, and severe. GSOTM complements this protocol with four camera-motion degradation settings: motion blur (MB), rolling shutter (RS), their combination (MB+RS), and pose noise (PN). Every method receives the same degraded posed observations. Neither sharp ground-truth images nor matched high-quality reference images are available during view allocation, reconstruction, repair selection, or distillation; sharp targets are used only for final reference-based evaluation.

Effect of Reliability-aware View Allocation

We isolate the reconstruction stage by comparing all-view $\mathcal{G}_0$ with RVA-selected $\mathcal{G}_0$. As shown in Figure 3, all-view supervision propagates blur into thin structures and object boundaries, while RVA yields sharper contours and more stable local appearance. The initial-scene rows in Table 3 show the same trend: RVA-selected $\mathcal{G}_0$ reduces LPIPS by 0.011, 0.013, and 0.015 and improves MUSIQ by 1.71, 2.49, and 4.11 from slight to severe blur. CLIP-IQA remains comparable, with an equal score under slight blur and a small gain under severe blur. These results show that RVA reduces blur-related degradation and provides a cleaner initialization for repair.

Quantitative Comparison

Table 3 reports perceptual results on GS2E and GSOTM, together with all-view and selected-view initial scenes to separate view allocation from repair. For a fair comparison, all methods use the same input sequences and are evaluated at the same observed input poses against the same sharp targets. Under this protocol, ClearGS improves over DiFiX3D on all GS2E blur levels, reducing LPIPS by 0.016, 0.030, and

Table 3: Perceptual comparison on GS2E and GSOTM at observed input poses. GS2E averages ten scenes per blur level, and GSOTM averages four scenes per degradation type. LPIPS uses sharp targets; CLIP-IQA and MUSIQ are no-reference metrics. Best and second-best results are highlighted.

Method	LPIPS↓								CLIP-IQA↑								MUSIQ↑							
	GS2E				GSOTM				GS2E				GSOTM				GS2E				GSOTM			
	Slight	Medium	Severe	MB	RS	MB+RS	PN	Slight	Medium	Severe	MB	RS	MB+RS	PN	Slight	Medium	Severe	MB	RS	MB+RS	PN			
All-view G_0	0.397	0.414	0.430	0.323	0.334	0.419	0.476	0.266	0.261	0.256	0.194	0.212	0.196	0.227	34.82	32.02	30.11	36.75	50.25	34.51	34.06			
Select-G G_0 (Ours)	0.386	0.401	0.415	0.285	0.333	0.393	0.472	0.266	0.256	0.261	0.201	0.211	0.187	0.224	36.53	34.51	34.22	42.31	50.55	39.46	35.42			
DiFiX3D (Wu et al. 2025a)	0.349	0.372	0.388	0.224	0.224	0.322	0.249	0.258	0.255	0.253	0.231	0.253	0.213	0.230	41.53	39.03	36.93	41.64	59.77	39.94	51.64			
GeoQuery (Cao et al. 2026)	0.350	0.373	0.389	0.231	0.224	0.332	0.249	0.256	0.253	0.252	0.230	0.250	0.216	0.231	41.33	38.70	37.20	41.86	59.50	40.02	51.27			
SyncFix (Li et al. 2026)	0.353	0.377	0.395	0.253	0.227	0.334	0.250	0.253	0.246	0.242	0.225	0.249	0.208	0.227	39.99	36.76	34.44	37.90	59.00	35.67	50.26			
ClearGS (Ours)	0.333	0.342	0.350	0.193	0.225	0.302	0.197	0.288	0.287	0.293	0.314	0.296	0.243	0.267	44.67	43.57	43.18	62.03	64.25	52.64	59.07			

Table 4: Ablation of RVA coverage allocation. S/M/V are GS2E blur levels; MB/RS/MB+RS/PN are GSOTM degradation settings.

Variant	PSNR↑							SSIM↑							LPIPS↓						
	GS2E			GSOTM				GS2E			GSOTM				GS2E			GSOTM			
	S	M	V	MB	RS	MB+RS	PN	S	M	V	MB	RS	MB+RS	PN	S	M	V	MB	RS	MB+RS	PN
RVA w/o cov.	20.123	19.538	19.499	21.621	18.010	17.624	18.413	0.607	0.592	0.589	0.730	0.530	0.534	0.573	0.388	0.413	0.415	0.290	0.339	0.387	0.472
Full RVA	19.934	20.076	20.011	22.545	18.124	18.010	18.458	0.606	0.609	0.608	0.739	0.530	0.541	0.573	0.386	0.401	0.415	0.285	0.333	0.393	0.472

Table 5: Ablation of RIVR candidate selection. “w/o Obs.” replaces the raw video observation O_t with the current render R_t as the input to Restormer, while “w/o F_t ” removes the high-frequency fused candidate. The remaining repair-distillation pipeline is unchanged.

Metric	Variant	GS2E			GSOTM			
		S	M	V	MB	RS	MB+RS	PN
LPIPS↓	w/o Obs.	0.385	0.401	0.414	0.247	0.294	0.358	0.398
	w/o F_t	0.336	0.348	0.365	0.171	0.213	0.298	0.196
	Full RIVR	0.333	0.342	0.350	0.193	0.225	0.302	0.197
CLIP-IQA↑	w/o Obs.	0.283	0.270	0.279	0.217	0.239	0.209	0.258
	w/o F_t	0.280	0.275	0.282	0.263	0.280	0.221	0.269
	Full RIVR	0.288	0.287	0.293	0.314	0.296	0.243	0.267
MUSIQ↑	w/o Obs.	37.15	35.59	34.98	47.37	57.30	46.20	47.13
	w/o F_t	43.43	42.12	40.67	51.64	63.30	45.53	58.98
	Full RIVR	44.67	43.57	43.18	62.03	64.25	52.64	59.07

0.038 and increasing CLIP-IQA by 0.030, 0.032, and 0.040 from slight to severe blur. MUSIQ gains also grow with blur strength, from 3.14 to 6.25. On GSOTM, ClearGS achieves the best CLIP-IQA and MUSIQ in all four settings, and its LPIPS is best in three settings and within 0.001 of the best score under rolling shutter.

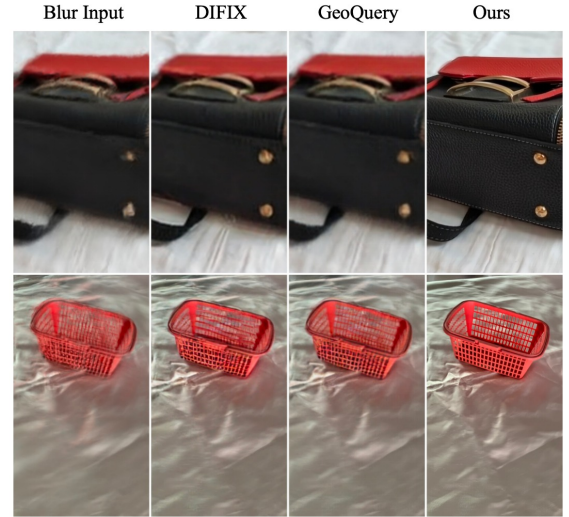


Figure 4: Qualitative comparison of repair outputs on two GS2E scenes. Each row shows the same camera pose. From left to right, the panels show the common pre-repair 3DGS rendering, the repairs produced by DiFix3D and GeoQuery, and the repair selected by ClearGS.

Qualitative Comparison

Figure 4 compares repair outputs at the same observed camera poses, while Figure 5 compares the final renderings with the input 3DGS, existing restoration baselines, and sharp ground truth. The pre-repair renderings retain visible blur

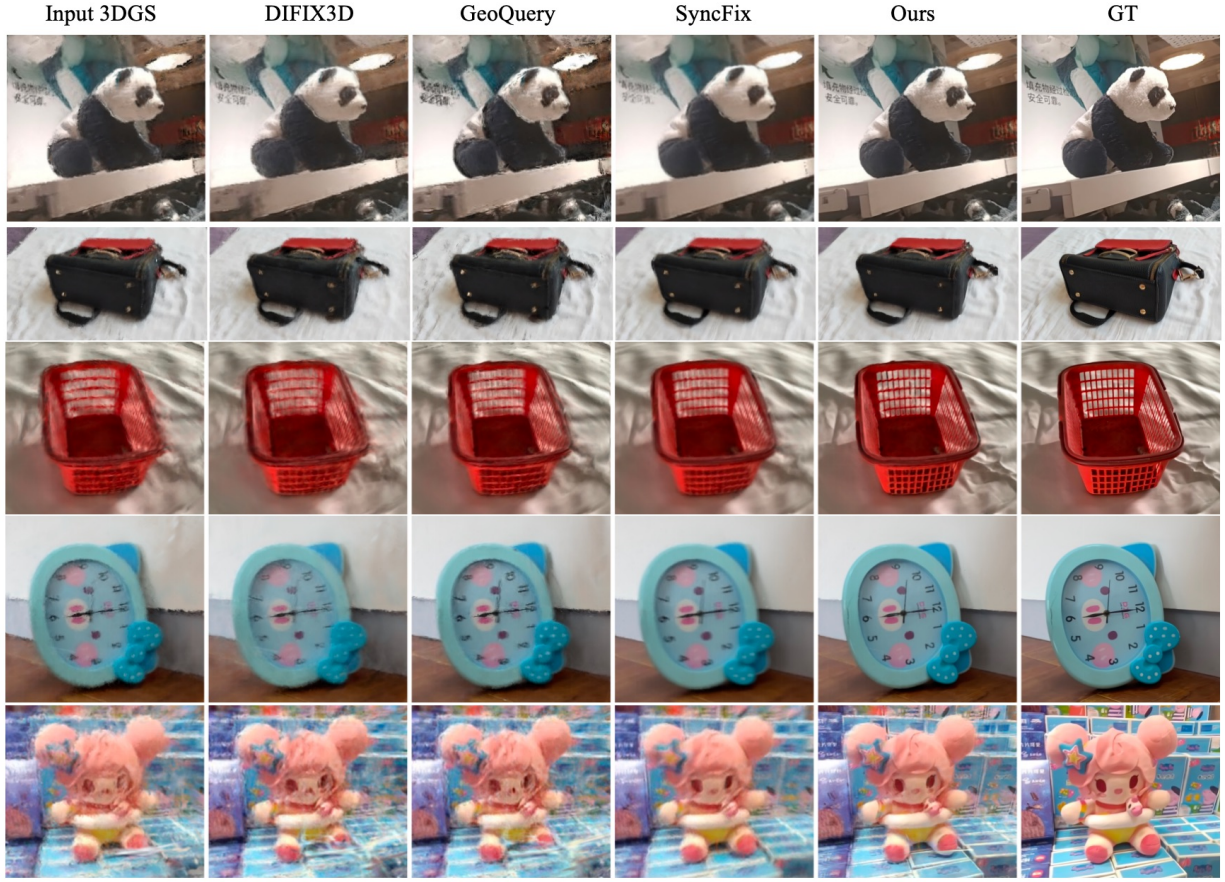


Figure 5: Qualitative comparison on five GS2E scenes. From left to right: the input 3DGS, DiFiX3D, GeoQuery, SyncFix, ClearGS, and the ground truth. ClearGS recovers sharper object boundaries and local details while preserving the scene appearance.

and weakened local structures, and the restoration baselines improve sharpness but can still leave residual artifacts or less stable scene details. In contrast, ClearGS produces cleaner object boundaries and more coherent local textures in the shown examples. These visual differences align with the quantitative results in Table 3, where ClearGS obtains the best overall perceptual scores and larger gains under stronger degradations.

Ablation Study

Coverage Allocation in RVA Table 4 evaluates the coverage top-up step in RVA. On GS2E, full RVA improves PSNR by 0.538 and 0.512 and SSIM by 0.017 and 0.019 under medium and severe blur, where reliability weighting is more likely to suppress useful but degraded trajectory regions. It also lowers LPIPS under slight and medium blur, while matching the no-top-up variant under severe blur. Under slight blur, the small PSNR drop and nearly unchanged SSIM suggest limited gain when most views remain reliable. On GSOTM, the effect is mixed but improves several settings, including RS and PN. These results show that coverage top-up complements reliability weighting by recovering

useful trajectory evidence under stronger degradation.

RIVR Candidate Selection Table 5 ablates the RIVR candidate pool. The full model selects among the current render R_t , the no-reference Restormer output from the aligned video observation O_t , and the high-frequency fusion candidate F_t . In w/o O_t , Restormer is still used, but its input is changed from O_t to R_t ; thus the ablation removes video evidence rather than the repair module. Removing this candidate consistently hurts LPIPS, CLIP-IQA, and MUSIQ on both datasets, with large MUSIQ drops on GSOTM motion blur and pose noise (14.66 and 11.94). Removing F_t generally lowers MUSIQ and CLIP-IQA, indicating that the fused candidate contributes perceptually preferred high-frequency details.

Conclusion and Discussion

We presented ClearGS for 3DGS reconstruction from handheld videos with mixed frame quality and uneven trajectory coverage. ClearGS assigns graded supervision through RVA, separating visual reliability, degradation risk, and geometric utility while preserving coverage in weakly observed trajectory regions. It then performs no-reference in-video restora-

tion: the current render serves as a pose-aligned structural anchor, and the corresponding raw video frame provides the evidence to be restored, without requiring matched clean references. A final consolidation pass revisits accepted repairs across the trajectory to preserve details learned early. Experiments on GS2E and GSOTM show improved perceptual quality, sharper boundaries, and more stable textures. Future work will study degradation-specific restoration and geometric checks for repaired evidence.

References

- Bai, S.; Cai, Y.; Chen, R.; Chen, K.; Chen, X.; Cheng, Z.; Deng, L.; Ding, W.; Gao, C.; Ge, C.; et al. 2025. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*.
- Cao, X.; Li, Y.; Zhang, Y.; Song, J.; Yan, C.; Li, W.; and Duan, L. 2026. GeoQuery: Geometry-Query Diffusion for Sparse-View Reconstruction. *arXiv preprint arXiv:2605.12399*.
- Chen, K.; Zhong, Y.; Li, Z.; Lin, J.; Chen, Y.; Qin, M.; and Wang, H. 2026a. Quantifying and alleviating co-adaptation in sparse-view 3d gaussian splatting. *Advances in Neural Information Processing Systems*, 38: 115939–115968.
- Chen, T.; Dai, A.; Adang, M.; Gao, G.; and Schwager, M. 2026b. Coverage Optimization for Camera View Selection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 19443–19451.
- Chung, J.; Oh, J.; and Lee, K. M. 2024. Depth-regularized optimization for 3d gaussian splatting in few-shot images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 811–820.
- Jiang, C.; Chen, Y.; Zhang, Q.; Song, J.; Xu, S.; Yeung, D.-Y.; and Deng, J. 2026. FreeScale: Scaling 3D Scenes via Certainty-Aware Free-View Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 330–340.
- Ke, J.; Wang, Q.; Wang, Y.; Milanfar, P.; and Yang, F. 2021. Musiq: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, 5148–5157.
- Kerbl, B.; Kopanas, G.; Leimkühler, T.; Drettakis, G.; et al. 2023. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4): 139–1.
- Lee, B.; Lee, H.; Sun, X.; Ali, U.; and Park, E. 2024. Deblurring 3d gaussian splatting. In *European Conference on Computer Vision*, 127–143. Springer.
- Lee, D.; Lee, M.; Shin, C.; and Lee, S. 2023a. Dp-nerf: Deblurred neural radiance field with physical scene priors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 12386–12396.
- Lee, D.; Oh, J.; Rim, J.; Cho, S.; and Lee, K. M. 2023b. Exblurf: Efficient radiance fields for extreme motion blurred images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 17639–17648.
- Lee, J.; Kim, D.; Lee, D.; Cho, S.; Lee, M.; Lee, W.; Kim, T.; Wee, D.; and Lee, S. 2025. Comogaussian: Continuous motion-aware gaussian splatting from motion-blurred images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 26415–26424.
- Li, D.; Yadav, A.; Peng, C.; Chellappa, R.; and Bhattad, A. 2026. SyncFix: Fixing 3D Reconstructions via Multi-View Synchronization. *arXiv preprint arXiv:2604.11797*.
- Li, J.; Zhang, J.; Bai, X.; Zheng, J.; Ning, X.; Zhou, J.; and Gu, L. 2024. Dngaussian: Optimizing sparse-view 3d gaussian radiance fields with global-local depth normalization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 20775–20785.
- Li, Y.; Feng, C.; Tang, Z.; Deng, K.; Yu, W.; Tian, Y.; and Yuan, L. 2025. GS2E: Gaussian Splatting is an Effective Data Generator for Event Stream Generation. In *Advances in Neural Information Processing Systems*, volume 38.
- Liu, X.; Zhou, C.; and Huang, S. 2024. 3dgs-enhancer: Enhancing unbounded 3d gaussian splatting with view-consistent 2d diffusion priors. *Advances in Neural Information Processing Systems*, 37: 133305–133327.
- Ma, L.; Li, X.; Liao, J.; Zhang, Q.; Wang, X.; Wang, J.; and Sander, P. V. 2022. Deblur-nerf: Neural radiance fields from blurry images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 12861–12870.
- Ni, J.; Chen, Y.; Yang, Z.; Liu, Y.; Lu, R.; Zhu, S.-C.; and Huang, S. 2025. G4Splat: Geometry-Guided Gaussian Splatting with Generative Prior. *arXiv preprint arXiv:2510.12099*.
- Park, H.; Ryu, G.; and Kim, W. 2025. Dropgaussian: Structural regularization for sparse-view gaussian splatting. In *Proceedings of the computer vision and pattern recognition conference*, 21600–21609.
- Peng, C.; Tang, Y.; Zhou, Y.; Wang, N.; Liu, X.; Li, D.; and Chellappa, R. 2024. Bags: Blur agnostic gaussian splatting through multi-scale kernel modeling. In *European Conference on Computer Vision*, 293–310. Springer.
- Polyzos, K. D.; Bacharis, A.; Madhuvarasu, S.; Papanikolopoulos, N.; and Javidi, T. 2025. Activeinitsplat: How active image selection helps gaussian splatting. *arXiv preprint arXiv:2503.06859*.
- Seiskari, O.; Ylilammi, J.; Kaatrasalo, V.; Rantalankila, P.; Turkulainen, M.; Kannala, J.; Rahtu, E.; and Solin, A. 2024. Gaussian splatting on the move: Blur and rolling shutter compensation for natural camera motion. In *European conference on computer vision*, 160–177. Springer.
- Shen, Y.; Yang, Y.; Zhou, K.; and Shao, T. 2026. Sparse-to-Complete: From Sparse Image Captures to Complete 3D Scenes. *arXiv preprint arXiv:2605.05664*.
- Song, M.; Lin, X.; Zhang, D.; Li, H.; Li, X.; Du, B.; and Qi, L. 2025. D²GS: Depth-and-Density Guided Gaussian Splatting for Stable and Accurate Sparse-View Reconstruction. *arXiv preprint arXiv:2510.08566*.
- Wang, P.; Zhao, L.; Ma, R.; and Liu, P. 2023. Bad-nerf: Bundle adjusted deblur neural radiance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 4170–4179.
- Wu, J. Z.; Zhang, Y.; Turki, H.; Ren, X.; Gao, J.; Shou, M. Z.; Fidler, S.; Gojcic, Z.; and Ling, H. 2025a. Difx3d+: Improving 3d reconstructions with single-step diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 26024–26035.

Wu, R.; Mildenhall, B.; Henzler, P.; Park, K.; Gao, R.; Watson, D.; Srinivasan, P. P.; Verbin, D.; Barron, J. T.; Poole, B.; et al. 2024. Reconfusion: 3d reconstruction with diffusion priors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 21551–21561.

Wu, S.; Xu, C.; Huang, B.; Geiger, A.; and Chen, A. 2025b. Genfusion: Closing the loop between reconstruction and generation via videos. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, 6078–6088.

Xiong, H.; Muttukuru, S.; Upadhyay, R.; Chari, P.; and Kadambi, A. 2023. Sparsegs: Real-time 360 sparse view synthesis using gaussian splatting. *arXiv e-prints*, arXiv–2312.

Xu, W.; Gao, H.; Shen, S.; Peng, R.; Jiao, J.; and Wang, R. 2024. Mvpgs: Excavating multi-view priors for gaussian splatting from sparse input views. In *European Conference on Computer Vision*, 203–220. Springer.

Xue, S.; Dill, J.; Ahuja, D.; Dellaert, F.; Tsotras, P.; and Xu, D. 2026. Uncertainty-driven 3D Gaussian Splatting Active Mapping via Anisotropic Visibility Field. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 5014–5026.

Yin, X.; Zhang, Q.; Chang, J.; Feng, Y.; Fan, Q.; Yang, X.; Pun, C.-M.; Zhang, H.; and Cun, X. 2025. Gsfixer: Improving 3d gaussian splatting with reference-guided video diffusion priors. *arXiv preprint arXiv:2508.09667*.

Zamir, S. W.; Arora, A.; Khan, S.; Hayat, M.; Khan, F. S.; and Yang, M.-H. 2022. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 5728–5739.

Zhang, J.; Li, J.; Yu, X.; Huang, L.; Gu, L.; Zheng, J.; and Bai, X. 2024. Cor-gs: sparse-view 3d gaussian splatting via co-regularization. In *European conference on computer vision*, 335–352. Springer.

Zhang, Q.; Rozumny, D.; Girlanda, F.; Karaoglu, S.; Pollefeys, M.; Gevers, T.; and Oswald, M. R. 2026. Unblur-SLAM: Dense Neural SLAM for Blurry Inputs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 352–362.

Zhao, L.; Wang, P.; and Liu, P. 2024. Bad-gaussians: Bundle adjusted deblur gaussian splatting. In *European Conference on Computer Vision*, 233–250. Springer.

Zhu, L.; Narayana, M.; Stary, M.; Hutchcroft, W.; Wetzstein, G.; and Armeni, I. 2026. GaussFusion: Improving 3D Reconstruction in the Wild with A Geometry-Informed Video Generator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 15432–15442.

Zhu, Z.; Fan, Z.; Jiang, Y.; and Wang, Z. 2024. Fsgs: Real-time few-shot view synthesis using gaussian splatting. In *European conference on computer vision*, 145–163. Springer.